

## Documentation about React backend by Youssef Eslam

### Progress Update

- Started researching and learning about WebSockets and ROS nodes.
- Created Prototype #1 for the robot GUI communication system.
- Created a fake sensor publisher to simulate sensor readings without requiring the physical robot.
- Created a ROS subscriber node to receive the fake sensor data.
- Added data analysis to detect unsafe conditions such as high temperature and low distance.
- For the prototype, the sensors are temperature and distance for easier detection of unsafe conditions, but it will be changed.
- Connected the ROS backend to a WebSocket server.
- Connected the WebSocket server to the React frontend.
- Successfully displayed the received sensor readings on the React website.
- Added a connection status to show whether React is connected to the backend.
- Verified that the sensor values update dynamically on the React page.

### Conclusion

- I have established the basic communication architecture that can later be connected to the real sensors and robot hardware.
- Further code reference and explanation will be available after finishing
- Next will be adding control commands and an object detection model and connecting it to the front end

Connection: Connected

Temperature: 32 °C

Distance: 68 cm

Status: SAFE

```
youssef@youssef: ~/car/react - youssef@youssef: ~ — /usr/ x youssef@youssef: ~/car/react/: youssef@youssef: ~/car/react -
ature': 33, 'distance': 33, 'warning': 'SAFE'}
[INFO] [1787593233.728093811] [subscriber]: Sent: {'type': 'sensor', 'temper
ature': 33, 'distance': 70, 'warning': 'SAFE'}
[INFO] [1787593234.778323187] [subscriber]: Sent: {'type': 'sensor', 'temper
ature': 31, 'distance': 59, 'warning': 'SAFE'}
[INFO] [1787593235.741351159] [subscriber]: Sent: {'type': 'sensor', 'temper
ature': 25, 'distance': 97, 'warning': 'SAFE'}
[INFO] [1787593236.727905096] [subscriber]: Sent: {'type': 'sensor', 'temper
ature': 37, 'distance': 62, 'warning': 'HIGH TEMPERATURE'}
[INFO] [1787593237.729429921] [subscriber]: Sent: {'type': 'sensor', 'temper
ature': 38, 'distance': 72, 'warning': 'HIGH TEMPERATURE'}
[INFO] [1787593238.740310719] [subscriber]: Sent: {'type': 'sensor', 'temper
ature': 23, 'distance': 74, 'warning': 'SAFE'}
[INFO] [1787593239.727478362] [subscriber]: Sent: {'type': 'sensor', 'temper
ature': 38, 'distance': 49, 'warning': 'HIGH TEMPERATURE'}
[INFO] [1787593240.727657447] [subscriber]: Sent: {'type': 'sensor', 'temper
ature': 40, 'distance': 95, 'warning': 'HIGH TEMPERATURE'}
[INFO] [1787593241.732207556] [subscriber]: Sent: {'type': 'sensor', 'temper
ature': 33, 'distance': 13, 'warning': 'TOO CLOSE'}
```

```
youssef@youssef: ~/car/react — /usr/bin/python3 /opt/ros/lyrical/bin/ros2 run robot_nodes ...
~/car/react
youssef@youssef: ~/car/re: x youssef@youssef: ~ — /usr/bin/ youssef@youssef: ~/car/react/: youssef@youssef: ~/car/react -
[INFO] [1787593216.727043070] [fake_publisher]: Published: 40,41
[INFO] [1787593217.736931291] [fake_publisher]: Published: 33,52
[INFO] [1787593218.731680541] [fake_publisher]: Published: 31,24
[INFO] [1787593219.726908100] [fake_publisher]: Published: 32,68
[INFO] [1787593220.727327973] [fake_publisher]: Published: 36,94
[INFO] [1787593221.764948664] [fake_publisher]: Published: 24,89
[INFO] [1787593222.729604336] [fake_publisher]: Published: 37,55
[INFO] [1787593223.726892171] [fake_publisher]: Published: 27,41
[INFO] [1787593224.727807248] [fake_publisher]: Published: 31,46
[INFO] [1787593225.727872231] [fake_publisher]: Published: 27,96
[INFO] [1787593226.728558152] [fake_publisher]: Published: 38,100
[INFO] [1787593227.727909351] [fake_publisher]: Published: 39,41
[INFO] [1787593228.727901949] [fake_publisher]: Published: 35,92
[INFO] [1787593229.734004162] [fake_publisher]: Published: 22,76
[INFO] [1787593230.728312828] [fake_publisher]: Published: 33,29
[INFO] [1787593231.729605033] [fake_publisher]: Published: 38,78
[INFO] [1787593232.729937609] [fake_publisher]: Published: 33,33
[INFO] [1787593233.727716607] [fake_publisher]: Published: 33,70
[INFO] [1787593234.777793160] [fake_publisher]: Published: 31,59
```